

Question	Answer	Mark	Mark scheme
21 (a)	$\frac{1}{4}$	M1	for $f(1) = 3 \times 1^2 + 1 (= 4)$ and a clear intention to find $g(“4”)$ or for $\frac{4}{(3 \times 1^2 + 1)^2}$ or for stating $gf(x)$, eg $\frac{4}{(3x^2 + 1)^2}$ oe
(b)	$\sqrt[4]{\frac{48}{x-1}}$	A1	oe
		M1	for finding $fg(x)$, eg $3 \times \left(\frac{4}{x^2}\right)^2 + 1$ or $\frac{48}{x^4} + 1$
		M1	for start of method to find the inverse of $fg(x)$, eg $y - 1 = 3 \times \left(\frac{4}{x^2}\right)^2$ or $y - 1 = \frac{48}{x^4}$ or $x - 1 = \frac{48}{y^4}$ or $x - 1 = 3 \times \left(\frac{4}{y^2}\right)^2$
		M1	for $y^4 = \frac{48}{x-1}$ or $x^4 = \frac{48}{y-1}$ or for a final answer of $\sqrt[4]{\frac{48}{y-1}}$
		A1	oe